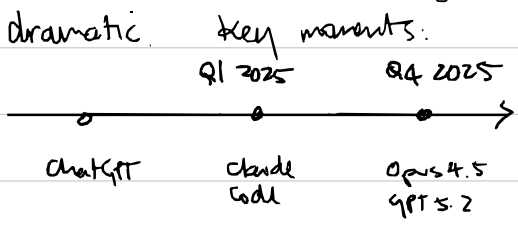
Notes for seminar on AI agents for physicists

Start with why:

LLMs have been improving dramatic. Key moments:



Insert GPT 5.2 headline on Physics

- If you experimented with chat interface in Q1 2025 you may well have dismissed LLMs as being useful for anything but small subset of tasks

- (Insert here the "maximally unreliable" comments LLMs Lie, Cheat, Steal ... Fawn & Flatter)

→ Slightly deranged MSc student

- I am not sure what you have experienced, but by Q4 2025 LLMs had become genuinely useful for nearly all knowledge work.

BUT to enjoy this you must know a couple of things

1. LLM requires deliberate practice
   It is an alien tech requiring a completely different way of working/reasoning
   You won't be good straight away
2. You probably need to pay $
3. You need to install Claude Code (or Cline / etc.)

With correct workflow you can truly reliably automate many scientific processes. E.g. it is now possible to fully automatically reproduce papers.

This is all possible with basic tooling. Absolutely no need for any additional subscriptions to Anthropic, Microsoft, ...

We are entering era of hyper-personalised software.

Cheaper + Easier (cognitively) to Build it yourself!

But FIRST, as a scientifically trained audience you will likely appreciate an explanation of how LLMs work → This will explain their strange behaviours

Section 1

Most important concepts.

1. LLM as stateless nondeterministic function
2. Context window
3. Context rot
4. Tool use.

→ Demo with python script